

Task 1: Product Management REST API

1. Task Objective

Develop a **Product Management REST API** using ASP.NET Core Web API and Entity Framework Core.

The API must support:

- Product Model creation
- SQL Server database integration
- Entity Framework Core Code First
- Migration and database creation
- DTOs for API requests/responses
- CRUD API endpoints
- Additional filtering/search endpoints
- FluentValidation
- Common API response format
- Proper HTTP status codes
- Error handling through common response

2. Product Information

Field	DataType	Description
ProductId	int	Primary Key
ProductName	varchar(100)	Product name
ProductCode	varchar(50)	Unique product code
Category	varchar(50)	Product category
Price	decimal(18,2)	Product price
StockQuantity	int	Available stock
ManufactureDate	datetime	Manufacturing date
ExpiryDate	datetime	Product expiry date
IsAvailable	bit	Product availability

3. API Endpoints

Method	Endpoint	Purpose
POST	/api/products	Add new product
GET	/api/products	Get all products
GET	/api/products/{id}	Get product by ID
PUT	/api/products/{id}	Update product
DELETE	/api/products/{id}	Delete product
GET	/api/products/available	Get available products
GET	/api/products/price-range?min=500&max=5000	Products within price range
GET	/api/products/summary	Total products, available products and average price

4. FluentValidation Requirements

Implement **minimum 10 validation rules**.

#	Field	Validation Requirement
1	ProductName	Required, Minimum 3 and maximum 100 characters, Must not contain leading/trailing whitespace
2	ProductCode	Required, Must not contain whitespace
3	Category	Required
4	Price	Must be greater than 0, Must not exceed ₹10,00,000
5	StockQuantity	Must be greater than or equal to 0
6	ManufactureDate	Must not be a future date
7	ExpiryDate	Must be greater than ManufactureDate
8	IsAvailable	Must contain a valid Boolean value

5. Common API Response

Every endpoint must return:

Status
Message
Data
Error

6. Expected HTTP Status Codes

Scenario	Status Code
Successful GET	200 OK
Successful POST	201 Created
Successful PUT	200 OK
Successful DELETE	200 OK / 204 No Content
Validation Failure	400 Bad Request
Product Not Found	404 Not Found
Unexpected Server Error	500 Internal Server Error